

Direct and Indirect Effects of Investment Tax Incentives[†]

By ADRIAN LERCHE*

This paper estimates the direct effects and indirect spillover effects of investment tax credits on firms. Exploiting a differential tax credit rate change by firm size in the German manufacturing sector, I find that lowering a firm's investment cost by 7.6 percent increases its capital stock by 17.7 percent and employment by 12.0 percent. Positive local spillovers generate one additional manufacturing job for each directly created job, are strongest between firms in industries connected through input-output linkages, and arise within distances of five kilometers. Firms dependent on local consumer demand also increase employment, while within-industry spillovers generate small negative effects. (JEL D22, H25, H32, J23, L25, L60, R11)

With the accumulation of capital commonly seen as key to economic growth, governments have long used investment tax credits and similar tax incentives to reduce the cost of capital. While recent evidence suggests that such policies can increase capital investment at the firm level (Zwick and Mahon 2017; Maffini, Xing, and Devereux 2019), general equilibrium spillovers determine how these direct effects translate into outcomes at aggregate levels (Guren et al. 2021). For example, agglomeration economies and local demand effects may multiply the direct firm effects. In contrast, fixed labor supply and product-market competition may lead to shifts in production across firms with unequal tax incentives, thereby offsetting the direct effects and possibly creating efficiency loss due to the misallocation of productive factors (Hsieh and Klenow 2009; Garicano, Lelarge, and Van Reenen 2016).¹

This paper explores the effectiveness of investment tax incentives in promoting firm production by estimating not only the direct effects of lowering the investment cost but also indirect effects arising from general equilibrium spillovers. To do so,

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¹Tax policies often provide better tax provisions for small firms (Maffini, Xing, and Devereux 2019; Benzarti and Harju 2021; Moon 2022) or, as with accelerated depreciation policies, implicitly favor specific industries (Zwick and Mahon 2017). Similarly, place-based tax policies target firms within particular regions (Slattery and Zidar 2020).

I focus on an investment tax credit policy introduced after German reunification. This policy—with annual tax expenditures of around €1–2 billion (\$1.2–2.4 billion)—was one of the leading support programs for firms in former East Germany to mitigate the considerable economic differences that had developed during the division of Germany.²

Investment tax credits allow firms to deduct a share of their cost of investment purchases from taxable income. A 1999 policy change increased the deduction rate on a wide range of equipment for small manufacturing firms with up to 250 employees from 10 to 25 percent, while increasing the rate for large manufacturing firms with more than 250 employees from 5 to just 12.5 percent. With small firms recouping an additional 15 cents of every dollar of investment and with large firms recouping only an additional 7.5 cents, the policy change implied a larger reduction in the user cost of capital for small firms. I exploit this differential change to identify causal effects.

In the first part of the empirical analysis, using a difference-in-differences design that compares the response of firms below and above the 250-employee firm-size cutoff (excluding firms immediately around the cutoff), I estimate the direct (partial equilibrium) effects of the policy. This analysis shows that investment tax credits have broad positive effects on firm outcomes. Small manufacturing firms increase their capital stock by 16.3 log points (17.7 percent) and employment by 11.3 log points (12.0 percent) relative to large firms. The effect on investment amounts to 23.0 log points (25.9 percent). Moreover, a detailed analysis of employment by skill and occupation reveals positive effects for all worker groups considered.

An important identification challenge is that unrelated shocks might drive the differential production decisions of small and large firms. To alleviate this concern, I include industry-year and region-year fixed effects in all empirical specifications. When I further control for pretreatment trends or estimate the dynamic effects, I do not find evidence for differential pretreatment behavior. The results remain unaffected when I control for heterogeneous firm responses along continuous measures of firm size. I also show that firms reacted to the policy by bunching right below the 250-employee cutoff.

Since firms do not operate in isolation, they may be affected independently of their investment tax incentives received through general equilibrium spillovers. These spillovers arise when the policy influences the aggregate economic environment. At the county level, by scaling up their production, directly affected firms may increase the demand for intermediate inputs, workers' wages, and household income. In counties with relatively more small manufacturing firms, the aggregate direct effects of the policy will be stronger, thereby generating more scope for local spillover effects. I rely on this relationship by comparing the response of firms by the employment share of small firms in the manufacturing sector in a given county (excluding a firm's own employment). Combining both difference-in-differences designs allows me to determine the local indirect effects of the policy change on firms while simultaneously controlling for the direct effects.

²In the United States, investment tax credits played a prominent role at the national level between 1962 and 1986, with tax expenditures totaling \$21 billion in 1985 (Chirinko 2000). At the state level, 40 percent of states in the United States offered investment tax credits in 2004 (Chirinko and Wilson 2008).

I find a substantial positive impact of local spillovers on firm production. For the average East German manufacturing firm, the larger reduction in the cost of capital for small firms leads to a 15.7 log point (17.0 percent) increase in the capital stock and a 6.4 log point (6.6 percent) increase in employment through local spillovers. Accounting for these spillovers increases on average the effects on capital from 9.7 to 25.4 log points and on employment from 6.3 to 12.7 log points. These results are robust to including a wide set of county and industry characteristics interacted with year fixed effects, do not depend on the sample selection, and remain unaffected when I control for changes in a concurrent support program for firms. Furthermore, a placebo analysis in which I select comparable manufacturing firms in West Germany yields effects close to zero.

To learn about the source of the indirect effects, I analyze the heterogeneous responses of manufacturing firms. The results show that economic closeness between firms is important, with the local spillover effects being strongest between firms in industries connected through input-output linkages. In particular, the spillovers primarily arise from firms in downstream industries that likely increase their demand for local intermediate inputs. At the same time, I find that local wages increase, pointing to imperfectly elastic local labor supply. However, the impact on the production cost is not strong enough to offset the positive spillover effects.

The empirical analysis relies on extensive administrative surveys and matched employer-employee data. Through firm address information, I am able to estimate the strength of the spillover effects for flexible geographic distances between firms. The analysis shows that the effects are highly localized, occurring for firms with distances of up to five kilometers (approximately three miles).

Beyond the local manufacturing sector, firms dependent on local consumer demand, such as those in the retail sector, experience positive spillovers, while for business service firms, they are negative. For West German manufacturing firms, spillovers within narrowly defined industries lead to a decrease in employment although this effect is primarily driven by smaller firms. In line with this evidence, an analysis of aggregate effects at the county-industry level shows that local changes in employment are driven by the manufacturing sector and that the overall effect on the West German manufacturing sector is limited. An analysis of worker flows reveals that initially nonemployed workers finding a job and employed workers at risk of nonemployment keeping their job explain a sizable share of the employment effect in the manufacturing sector. While the estimated effects cannot quantify all general equilibrium channels relevant for aggregate outcomes (see Nakamura and Steinsson 2014; Beraja, Hurst, and Ospina 2019; Andersen et al. 2022), the results provide useful input into the understanding of the propagation of firm-level shocks.

This paper contributes to studies that exploit crosssectional tax policy variation for investigating the investment responses of firms (Cummins, Hassett, and Hubbard 1994; Yagan 2015; Zwick and Mahon 2017; Maffini, Xing, and Devereux 2019; Liu and Mao 2019; Moon 2022).³ Recent studies have expanded the analysis to a wider set of firm outcomes. For US accelerated depreciation policies, Tuzel

³Important earlier contributions on the investment effects of tax policy are Hall and Jorgenson (1967); Abel (1980); Summers (1981); Hayashi (1982); Chirinko, Fazzari, and Meyer (1999); and Desai and Goolsbee (2004).

and Zhang (2021) find no effect on total employment but a shift toward skilled workers, while Curtis et al. (2021) find positive effects on employment, capital, and sales. I provide novel evidence of these direct effects for a prominent policy in Germany.

I build on this evidence to connect to studies that examine tax policy at aggregate levels (House and Shapiro 2008; Romer and Romer 2010; Mertens and Ravn 2013; Suárez Serrato and Zidar 2016; Ohn 2019). Most closely related, Garrett, Ohn, and Suárez Serrato (2020) find positive effects of US bonus depreciation on local labor markets. I complement these studies by linking firm-level evidence with findings at aggregate levels in a unified empirical framework. This analysis disentangles the adjustment mechanisms that shape production at higher levels of aggregation and identifies local spillover effects as an important channel for the overall effectiveness of investment tax incentives. I also shed light on reallocation patterns of production across firms as another important factor in the aggregate response.⁴

This paper thus contributes more generally to the economic literature that empirically explores general equilibrium spillovers. Agglomeration economies are central to the understanding of localized production and urbanization.⁵ Yet given the difficulty of finding appropriate research settings, few studies have exploited quasiexperimental variation to identify these effects. At the firm level, Greenstone, Hornbeck, and Moretti (2010) and Gathmann, Helm, and Schönberg (2020) take advantage of large firm openings and mass layoffs, respectively, to explore the impact on firm production in affected regions. Huber (2018) provides evidence from lending cuts during the Great Recession. Exploiting variation from tax policy, I analyze the size, source, and dynamics of local spillovers in a setting that has not yet been studied in this context. Moreover, by analyzing agglomeration effects not only within administrative boundaries but also with respect to the distance between firms, I provide new evidence on the geographic scope of agglomeration effects (Rosenthal and Strange 2003; Arzaghi and Henderson 2008; Ahlfeldt et al. 2015).⁶

The paper proceeds as follows. Section I presents the investment tax credit policy and the variation I use in the empirical analysis. Section II provides an overview of the data, sample selection, and descriptive evidence of the firm behavior over time. Sections III, IV, and V present estimation results for the direct effects, local indirect effects on the manufacturing sector, and indirect effects on other firms in the economy, respectively. Section VI puts the results into context, and Section VII concludes.

⁴Giroud and Rauh (2019) show the importance of reallocation within firms across state borders for explaining the aggregate effects of state taxation on business activity.

⁵See, for example, Ciccone and Hall (1996); Glaeser and Maré (2001); Ellison, Glaeser, and Kerr (2010); and De la Roca and Puga (2017). Combes and Gobillon (2015) provide an overview of empirical strategies and the challenges involved in identifying agglomeration effects.

⁶Given the importance of local input-output linkages, this evidence is informative about the geography of production networks more generally (Bernard, Moxnes, and Saito 2019). The regional focus of the German investment tax credits further relates this paper to the place-based policy literature (see Glaeser and Gottlieb 2008; Kline and Moretti 2014; Neumark and Simpson 2015). Such policies commonly combine a mix of incentives and allocate funds at the discretion of local agencies, making direct comparisons of effects difficult.

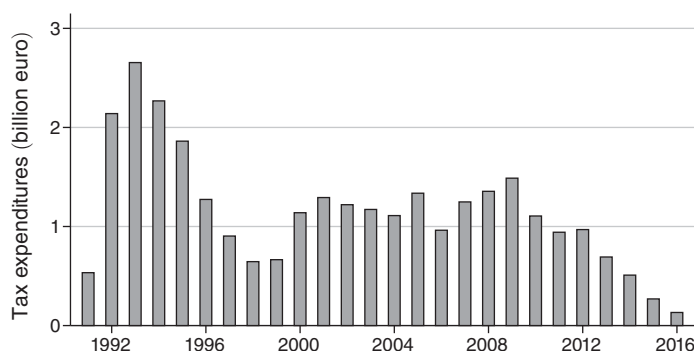


FIGURE 1. REPORTED TAX EXPENDITURES FOR INVESTMENT TAX CREDITS IN GERMANY

Notes: Data collected from the German government subsidy and tax relief reports (Supplemental Appendix F.e). For 1991–1998, tax expenditures are converted to euros. When several reports provide information for the same year, I select the value from the most recent report.

I. The German Investment Tax Credit Policy

A. Institutional Details

In 1990, Germany garnered worldwide attention by reuniting West and East Germany (I also use these terms to refer to the respective regions after reunification). However, stark differences in their political and economic development throughout their separation led to sizable regional inequality. In 1991, East Germany had 43 percent of the GDP per capita of West Germany, 46 percent of the capital stock per worker, 57 percent of the earnings per worker, and an unemployment rate of 9.5 percent, compared to 5.8 percent in West Germany (Supplemental Appendix F.a–F.d). To mitigate these economic differences, the German government provided considerable financial support. The investment tax credit program (*Investitionszulagengesetz*) was implemented in 1991, offering tax credits to firms in East Germany for equipment and structures investment. While at the beginning firms in all industries were eligible, by 1997 the program almost exclusively targeted the manufacturing sector.⁷

Tax credits were generous, with rates ranging from 5.0 to 27.5 percent. A firm eligible for the highest rate could, for example, claim a €275,000 (\$330,000) tax deduction for an investment of €1 million (\$1.2 million). Although tax credits usually cannot exceed tax liabilities, tax credits in the German program were refundable, meaning that firms received payments at the end of the business year irrespective of their tax liabilities.

Figure 1 summarizes the tax expenditures reported for the investment tax credit policy. From 1992 to 1995, they amounted to roughly €2 billion per year (\$2.4 billion). The annual cost declined to €600 million (\$720 million) by 1999, reflecting the

⁷The retail industry continued to have limited eligibility until 2001. Manufacturing-related business services, such as industrial design services or laboratories, gained access to tax credits in 1999. Hotels and other accommodation services became eligible in 2007.

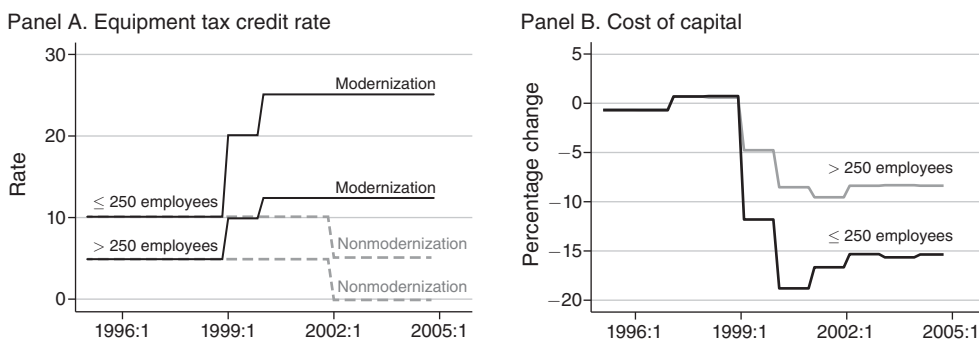


FIGURE 2. EQUIPMENT TAX CREDIT RATE AND THE COST OF CAPITAL

Notes: Panel A presents the equipment tax credit rate by firm size. The tax credit rates apply to manufacturing firms in East Germany, excluding Berlin, and are the standard rates of the official government act for investment projects completed within a business year. Panel B shows the percentage change in the average user cost of capital for small and large manufacturing firms in East Germany relative to the period before the policy change. Calculations are based on the tax term of the user cost of capital, with the specific tax credit rate rules and general tax policy changes for each firm taken into account.

restriction of industry eligibility. The tax expenditures fluctuated around €1 billion (\$1.2 billion) per year thereafter and faded after the program ended in 2013.

In the empirical analysis, I focus on manufacturing firms in East Germany, excluding those in Berlin because of differences in the eligibility criteria.⁸ For the selected firms, changes in the equipment tax credit rate occurred at the start of 1999. Although this change was announced in August 1997, disputes with the European Union delayed final approval until the end of 1998.⁹ Panel A of Figure 2 depicts the standard equipment tax credit rate for this subgroup for 1995–2004. Even before the policy change, tax credit rates differed by the policy-relevant firm size, defined as the headcount of all employees except vocational trainees at the start of a business year. Small manufacturing firms, with at most 250 employees, received a tax credit rate of 10 percent on all equipment investment, whereas large ones received only 5 percent.

In 1999, tax credit rates were raised for modernization investment, with small firms receiving a rate of 20 percent and large firms receiving 10 percent.¹⁰ One year later, these rates increased to 25 and 12.5 percent, respectively.¹¹ For nonmodernization equipment investment, rates remained constant during the policy change in 1999 and decreased equally by 5 percentage points for both small and large firms in 2002.

⁸Firms in Berlin generally had less favorable conditions, with incentives applying only to specific types of investment and investment tax credit rates being lower for small firms.

⁹Using a dynamic regression specification in the empirical analysis, I find no evidence for anticipatory behavior of firms.

¹⁰At the same time, an investment limit for receiving the higher tax credit rate of €2.56 million per year (\$3.07 billion) for firms below the employment cutoff was eliminated. While tax credits for investment in structures were introduced in 1999, the deduction rates did not differ by firm size.

¹¹Small manufacturing firms in regions close to Berlin were not eligible for the additional increase in the rate for modernization investment in 2000. Firms in regions close to the Polish or Czech borders received an additional rate of 2.5 percent for modernization investment in 2001–2009. For investment projects spanning several years, the exact start and end dates influenced the applicable tax credit rate.

The policy broadly defined modernization investment, including any equipment that could potentially increase production.¹² To quantify the share of modernization investment, I rely on two complementary approaches, both of which utilize information on claimed tax credits from confidential corporate tax records (Supplemental Appendix F.1–F.3). The first approach compares the total tax credit claims of manufacturing firms before and after the policy change, while the second one compares the effective tax credit rates of small and large East German manufacturing firms. The two analyses reveal a share of modernization investment in the range of 80–95 percent. I provide details and empirical results for these analyses in Supplemental Appendix A.1.

I determine the impact of the policy change on the investment cost of manufacturing firms for a share of modernization investment of 80 percent and by taking the specific investment tax credit rates for each firm into account. As the German tax credits did not reduce the depreciation base, I also take the changes of other tax policies into account. Importantly, none of their provisions varied by firm size. I describe the details of the German tax system and the steps for calculating the user cost of capital in Supplemental Appendix A.2. Panel B of Figure 2 plots the percentage change in the average cost of capital relative to the years before the policy change separately for small and large firms. For the period considered, the policy change decreased the cost of capital for small firms by 15.6 percent and for large firms by 8.0 percent.

B. *From Policy Variation to Identification Strategy*

The differential change in the incentives by firm size creates useful variation for the analysis of the investment tax credit policy. If manufacturing firms react to investment tax incentives, the larger reduction of the cost of capital for small firms should translate to a stronger production response of these firms.¹³ A comparison of the trends in production of small and large firms before and after the policy change therefore can identify the direct effects of investment tax credits.

Moreover, because small and large manufacturing firms are not equally distributed across space, the policy change creates regional variation in the average cost reduction. Any direct effects will therefore lead to a stronger aggregate manufacturing production response in counties with a higher share of small firms, creating a larger potential for local spillovers through, for example, the labor market, production networks, or consumer demand. A comparison of firms by the share of small firms in the manufacturing sector in a county before and after the policy change can therefore identify these indirect effects. To separate the direct effect on a particular manufacturing firm from the indirect effects generated by other firms in a region, I calculate this share for the surrounding firms in a county, excluding a firm's own employees. To account for the economic importance of a firm, I weight the share by employment. A firm production model that introduces region-specific

¹² See German Federal Ministry of Finance (2001) regulations published on June 28, 2001.

¹³ While the jump in the tax credit rate at the firm size cutoff might appear appealing for a regression discontinuity design, firms can actively change their employment over time and thus have an incentive to bunch at the cutoff. I examine this behavior in Section IIID.

productivity dependent on overall employment in a region, detailed in Supplemental Appendix B, can derive this estimation strategy more formally.

II. Data and Descriptive Evidence

A. Data and Sample

The empirical analysis relies mainly on two comprehensive administrative data sources, the *AFiD-Panel Industriebetriebe* (AFiD) from the German Federal Statistical Office (Supplemental Appendix F.4) and matched employer-employee data from the *Beschäftigten-Historik* (BeH) from the Institute for Employment Research (Supplemental Appendix F.5 and F.6).¹⁴

The AFiD dataset, which covers the universe of manufacturing and mining firms with at least 20 employees, is particularly well suited for the general analysis of firm capital stock, employment, and production output. The data, collected through administrative surveys at the establishment level, form the basis for German official statistics. To align the analysis with the relevant level of the investment tax credits, I aggregate the data by using firm identifiers. To obtain capital stock measures, I rely on capital depreciation information from the supplementary *Kostenstrukturhebung* (Supplemental Appendix F.7) and combine this information with industry-level depreciation rates from Müller (2017) and firm investment from the AFiD data.¹⁵ Supplemental Appendix C describes the calculation of capital stock and other data processing steps in detail.

For an in-depth analysis of employment effects, I use the BeH data, which comprise the universe of employees covered by the German social security system.¹⁶ For a standard set of employment and wage outcomes, Ganzer et al. (2018, total population) provide aggregated data at the establishment level for June 30 of each year. I expand this dataset with information on employment by skill and occupation groups, worker flows, and the geographic addresses of establishments.

Given German data protection laws, the BeH data cannot be linked with the AFiD data. To make both datasets comparable, I aggregate the BeH data at the firm level by using firm names available through German Federal Employment Agency registries (Schäffler 2014; Supplemental Appendix F.8) and restrict the sample to firm observations with at least 20 employees. I further convert the daily average wage of full-time employees to monthly values. The AFiD data include the monthly wage bill, from which I obtain an average monthly wage by dividing by the number of employees (the full-time share of East German manufacturing employment is around 95 percent during the period considered). In both datasets, for multiestablishment firms I select industry and location according to the establishment with the highest number of workers over time and calculate the employment share of small

¹⁴For several analyses, I rely on further county and industry level data. Supplemental Appendix C.6 summarizes the variables and data sources (Supplemental Appendix F.14–F.28).

¹⁵The *Kostenstrukturhebung* is a yearly firm survey of a stratified random sample in the manufacturing and mining sector conducted by the German Federal Statistical Office.

¹⁶The German social security system covers roughly 80 percent of the total workforce. Civil servants, the self-employed, and military personnel are not included.

firms in the manufacturing sector in a given county by determining the shares for each establishment separately and weighting them by establishment employment.

The main analysis sample includes manufacturing firms in East Germany between 1995 and 2004. As 1995 is the first year with available AFiD data, it constitutes a natural starting point. By then, the privatization of state-owned firms was also nearly completed, and the convergence with West Germany had slowed down significantly despite remaining economic differences (Sinn 2002). After 2004, the firm size definition of the policy was changed, thus making a comparison with prior years difficult.¹⁷ I exclude firms in Berlin, where different tax credit rate rules applied, and restrict the sample to firms that were active throughout the sample period. I further select firms with a policy-relevant firm size in 1998 between 40 and 1,500 and exclude those with more than 225 employees and fewer than 275. I introduce these firm size restrictions to limit any undesired influence of greatly differently sized firms and the nonlinear incentives faced by firms close to the cutoff. Robustness tests show that the main results do not depend on this particular firm size selection.

Table 1 presents summary statistics for 1995–2004 for manufacturing firms in West and East Germany and for the analysis sample divided into small and large firms.¹⁸ Panel A, which shows statistics for the AFiD data, reveals clear differences between manufacturing firms in the West (column 1) and the East (column 2), supporting the well-documented finding that economic differences persisted after German reunification. For the analysis sample (columns 3–4), as expected, both groups show differences in average firm outcomes. In contrast, the employment share of small firms in the manufacturing sector does not depend much on the firm size groups. By multiplying average employment with the number of firm observations, the table also provides information on aggregate employment. Firms in the sample cover 38 percent of manufacturing employment in East Germany, with two thirds of them in small firms.

Panel B summarizes the BeH data. For each subgroup, mean employment and wages compare well with their counterparts from panel A, likely due to the high data collection standards for both datasets. Both datasets also determine a comparable share of multiestablishment firms. The alignment of this outcome reinforces the reliability of the firm identifiers generated from the BeH firm name data.

B. Trends in Manufacturing

For descriptive evidence of trends in the manufacturing sector, Figure 3 presents outcomes by year. Panels A and B show the distinct development of the aggregate capital stock and aggregate employment, respectively, for East and West Germany. The capital stock increased substantially more in East Germany, and East German manufacturing employment fluctuated around a stable trend, while West German manufacturing employment declined.¹⁹

¹⁷From 2005, the cutoff followed the definition of small and medium firms by the European Union, that is, it took the ownership structure into account and was defined jointly by the number of employees, sales, and total assets.

¹⁸Supplemental Appendix C.5 provides statistics of all key variables in the empirical analysis.

¹⁹These changes are relative to a capital stock of €630 billion in West Germany and €69 billion in East Germany, and employment of 5.7 million workers in West Germany and 567,000 workers in East Germany in 1995. As a related result, the share of manufacturing employment increased in East Germany while it declined in West Germany over the sample period.

TABLE 1—SUMMARY STATISTICS FOR MANUFACTURING FIRMS

	All manufacturing		Analysis sample	
	West (1)	East (2)	Small (3)	Large (4)
<i>Panel A. AFiD data</i>				
Capital (million)	29.02 (354.35)	21.17 (139.06)	12.61 (19.17)	132.16 (387.81)
Employees	168.52 (1,444.84)	91.28 (369.93)	93.94 (54.74)	517.56 (313.75)
Total sales (million)	35.45 (479.55)	13.31 (72.07)	11.42 (15.66)	101.05 (206.69)
Average monthly wage	2,451.70 (1,104.96)	1,680.30 (541.87)	1,720.48 (475.97)	2,220.89 (634.81)
Multiestablishment firms (percent)	9.26 (28.99)	5.16 (22.12)	3.82 (19.18)	15.19 (35.90)
Employment share small firms county (percent)	41.90 (13.10)	65.80 (16.32)	66.89 (15.65)	62.23 (17.02)
Observations	335,778	60,387	15,188	1,326
<i>Panel B. BeH data</i>				
Employees	141.32 (759.87)	77.00 (144.78)	91.65 (56.14)	470.77 (279.28)
Average monthly wage full-time	2,597.68 (744.18)	1,738.19 (555.63)	1,790.68 (508.65)	2,267.71 (678.56)
Share college educated (percent)	14.85 (26.74)	10.46 (11.89)	10.79 (10.51)	14.13 (10.97)
Multiestablishment firms (percent)	7.96 (27.06)	5.35 (22.50)	6.79 (25.16)	10.61 (30.81)
Employment share small firms county (percent)	46.63 (14.17)	73.86 (16.38)	74.88 (15.67)	69.67 (17.91)
Observations	380,582	71,272	18,960	1,630

Notes: This table provides the means with standard deviations in parentheses for selected outcomes of the AFiD data in panel A and the BeH data in panel B. Columns 1–2 include all observations of manufacturing firms for 1995–2004 for West Germany and East Germany, respectively, excluding Berlin. Columns 3–4 include all observations from the main analysis sample split into those of small firms with a policy-relevant firm size in 1998 of up to 225 employees (column 3) and large firms with at least 275 employees (column 4). In both panels, the numbers of observations refer to the employee information.

Panels C and D present trends in the average log capital stock and log employment, respectively, for firms in the main sample. For a closer match with the empirical analysis, I adjust both firm outcomes for differences across industry and federal states. To show developments both by a firm's size and by the surrounding manufacturing firms in a county, I further divide small and large firms according to the employment share of small firms in the manufacturing sector in a county. If spillovers affect firm production, then firms should behave differently not only by their size but also by the size of the firms surrounding them. To facilitate comparison, I adjust the pretreatment mean for each group to the overall sample mean.

For almost all the firm groups, capital and employment increased until 2000.²⁰ Given the sizable increase in tax incentives in 1999, the limited firm response may be explained by a low growth period, commonly attributed to the 1997 Asian financial

²⁰The balanced sample excludes entering and exiting firms, which drive the decline in aggregate manufacturing employment in East Germany through 1997, evident in panel B of Figure 3.

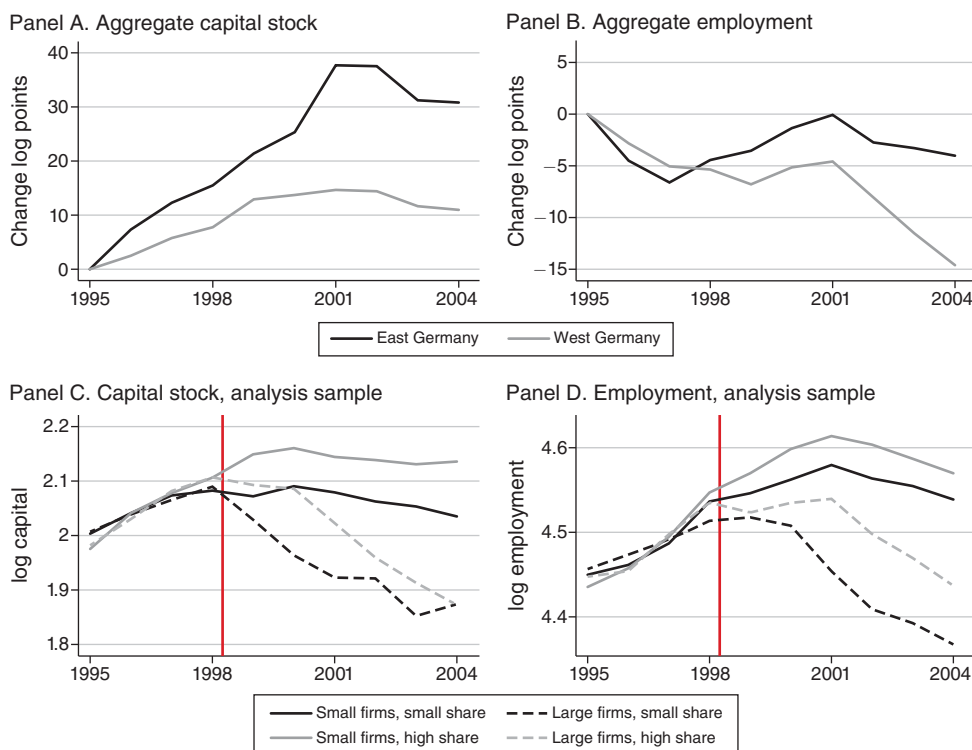


FIGURE 3. TRENDS IN THE GERMAN MANUFACTURING SECTOR

Notes: The figure plots the change relative to 1995 in the log aggregate value of the capital stock in panel A and the log aggregate employment in panel B separately for the sample of East and West German manufacturing firms, excluding Berlin. Panels C and D show the average log value of capital stock and average log employment, respectively, for four subgroups of the main sample over time. Values in each year are adjusted for differences across industries and federal states. The groups distinguish between small and large firms as defined in Table 1. Both groups are further subdivided by the employment share of small firms in the manufacturing sector in 1998 being below or above the median for large firms. The pretreatment group-specific means are set to the overall pretreatment sample means of each outcome.

crisis. A clear trend reversal occurred after 2000, when the dotcom crash led firms to contract. Both panels also show similar trends across firm groups until the policy change when the trends start to diverge. A comparison of the solid and dashed black lines shows that small manufacturing firms increased both their capital and employment relative to large firms after the policy change. A comparison of the black and gray lines instead shows the role of indirect effects. Firms surrounded by a high share of small firms outperform their counterparts surrounded by a low share, independently of their own reduction in investment costs.²¹ As is evident, firm production depends not only on a firm's own user cost of capital but also on the investment incentives received by other firms in the same county.

²¹ While the indirect effects appear smaller than the direct effects, the average share of small firms is 76 percent in the high-share group and 47 percent in the low-share group. Thus, even the low-share group benefits from local spillovers, with the comparison of both groups reflecting the difference in their spillover effects.

III. Direct Effects of Investment Tax Credits

The descriptive evidence provides a first indication for the importance of direct and indirect effects of investment tax credits. For a detailed analysis, this section starts by presenting results for the direct effects.

A. Direct Effect Specification

For the analysis of direct effects, I estimate the baseline regression specification

$$(1) \quad Outcome_{it} = \beta Small_{i,98} \times Post98_t + \psi_i + \psi_{nt} + \psi_{st} + \epsilon_{it},$$

where the dependent variable $Outcome_{it}$ represents different outcomes of firm i in year t . The variable $Small_{i,98}$ classifies firms according to their policy-relevant firm size in 1998 into small firms below the firm size cutoff and a comparison group of those above it. This variable is interacted with the variable $Post98_t$, which classifies the years after 1998 as the treatment period. This design exploits the difference in treatment intensity of small and large firms, with the parameter of interest β capturing the effect of the differential change in the cost of capital.

I add firm (ψ_i), industry-year (ψ_{nt} , 14 manufacturing groups), and federal state-year (ψ_{st}) fixed effects. The fixed effects can mitigate the concern that industry-specific shocks or region-specific policies and trends may coincide with the policy change. Furthermore, the firm production model highlights that fixed effects are important for controlling for differences in general equilibrium effects across geographic locations and industries, which may confound the direct effect analysis. The identifying assumption is that any differences in the indirect spillover effects within each fixed effect group are uncorrelated with the direct effects.

In robustness specifications, I interact county (X_c) and industry characteristics at the three-digit level (X_n) averaged over the pretreatment period with year fixed effects. As an important macroeconomic trend, the trade integration of Eastern European countries and China led to a 159 percent increase in German exports to these countries and a 164 percent increase in imports from these countries over the sample period.²² At the same time, Germany continued to struggle after reunification, experiencing sectoral employment changes that might have interacted with the policy. To control for these influences, I interact at the county and industry levels the net trade exposure to Eastern Europe and China and at the county level the shares of manufacturing, private services, and public services employment with year fixed effects.²³ I further include at the county level the unemployment rate, value added per worker, population density, total employment, and surface area, and at the industry level the high skill share, a regional concentration index, and an indicator for high innovation activity, all interacted with year fixed effects.

²²The AfID data show that West German manufacturing firms exported on average 15.3 percent of their sales in the pretreatment period, whereas East German manufacturing firms exported on average only 7.1 percent.

²³I follow Dauth, Findeisen, and Suedekum (2014) in the calculation of trade exposure.

TABLE 2—DIRECT EFFECTS OF INVESTMENT TAX CREDITS

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dependent variable: log capital</i>						
Small firm $\times$ post	0.163 (0.056)	0.155 (0.055)	0.172 (0.055)	0.154 (0.059)	0.166 (0.059)	0.151 (0.064)
Observations	15,146	15,128	15,128	15,128	15,128	15,128
<i>Dependent variable: log employment</i>						
Small firm $\times$ post	0.113 (0.035)	0.099 (0.027)	0.098 (0.028)	0.089 (0.028)	0.093 (0.030)	0.108 (0.033)
Observations	16,514	16,497	16,497	16,497	16,497	16,497
<i>Dependent variable: log domestic sales</i>						
Small firm $\times$ post	0.097 (0.044)	0.093 (0.041)	0.098 (0.041)	0.074 (0.042)	0.082 (0.043)	0.109 (0.049)
Observations	16,482	16,468	16,468	16,468	16,468	16,468
Pretreat. trends	No	Yes	Yes	Yes	Yes	Yes
$X_n \times$ year FE	No	No	Yes	No	Yes	Yes
$X_c \times$ year FE	No	No	Yes	No	No	No
County-year FE	No	No	No	Yes	Yes	Yes
Rank sales 1998 $\times$ post	No	No	No	No	No	Yes

Notes: The table reports estimates from regressions of specification (1). The baseline controls are firm, industry-year, and federal state-year fixed effects. Pretreatment trends are yearly pretreatment wage growth and employment growth interacted with time. The county controls X_c , interacted with year fixed effects, are the population density; total employment; surface area; unemployment rate; gross value added per worker; net trade exposure to East Europe and China; and the manufacturing, private services, and public services employment shares averaged for the pretreatment period. The industry controls at the three-digit level X_n , interacted with year fixed effects, are the high-skilled employment share, a regional concentration index, an indicator for high innovation activity, and the net trade exposure to East Europe and China averaged for the pretreatment period. The dependent variables are the log value of the capital stock, log employment, and log domestic sales. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

B. Main Direct Effect Results

Table 2 summarizes the direct effect results, with column 1 reporting estimates from baseline specification (1). Focusing first on capital and labor as important inputs to production, I find that lowering the user cost of capital leads small firms to increase their capital stock by 16.3 log points relative to large firms. Finding positive effects is consistent with the theoretical prediction that firms expand their use of capital because of both a substitution effect away from labor and a positive scale effect from reducing the overall production cost. The effect on employment amounts to an increase of 11.3 log points. This effect is smaller than that for capital, as would be expected from a substitution away from labor toward capital.²⁴ Given the expansion of capital and employment, I turn to the effect on output. I use domestic sales as the main output measure, as this outcome limits the influence of firms with volatile

²⁴From the capital and employment effects, the elasticity of substitution can be calculated as $\sigma = \frac{d \ln(K_i/L_i)}{d \ln(w/r)}$ = $(0.163 - 0.113)/0.076 = 0.658$. Studies commonly determine this elasticity to be below one at the firm level (Chirinko, Fazzari, and Meyer 2011; Raval 2019; Oberfield and Raval 2021). Large standard errors for the estimated elasticity limit a more detailed analysis.

export shares in the pretreatment period.²⁵ The estimated effect of 9.7 log points is close in magnitude but smaller than the input response.

Taken together, the results imply a scaling up of the production of directly affected firms. The positive labor demand response supports policymakers who promise to create more jobs through investment tax incentives. However, a comparison between the magnitudes of the input and output effects suggests that firms do not improve productivity and may even experience an efficiency loss. Investment tax credits seem to create limited incentives for firms to modernize their production technologies, casting doubt on the goal of policymakers to spur long-term growth, as firms may return to their prior production levels once benefits end.

I check the robustness of the baseline results to adding additional control variables. The validity of the difference-in-differences results relies on a parallel trends assumption: if the change in the investment tax incentives was equal, small firms would have the same trends in outcomes as large ones. To account for any differential trends from before the policy change, I interact the average yearly wage growth and employment growth of firms in the pretreatment period (log difference) as available proxies for changes in the productivity and the scale of production in both main datasets linearly with time. Column 2 shows that including these trends leaves the results largely unchanged. A benefit is that the average effects tend to be more precisely estimated. I further find that additionally including county and industry controls interacted with year fixed effects (column 3), as well as county-year fixed effects (columns 4–5), which can control for shocks to small regional units, does not meaningfully change the results.

Column 6 shows that the estimates do not materially change when I control for the firm size rank (based on total sales in 1998) interacted with the treatment period dummy. Previous studies suggest that small firms react particularly strongly to investment tax incentives due to, for example, financial constraints. If so, my research design would capture both the effect of differences in investment costs and the heterogeneous response of firms. I exploit the fact that the investment tax credit policy leads to a jump in the investment tax credit rate at the firm size cutoff, while the impact of any constraints should change continuously along firm size (with the functional form given by the firm size measure). The robustness of the results suggests that the investment tax credit policy did not induce heterogeneous responses by firm size, possibly because investment tax credits do not reduce financial constraints in the same way as other tax policies.²⁶

For evidence on the development of the effects over time, I implement for specification (1) an event-study framework, which interacts the firm size group variable with year dummies. Figure 4 shows for all outcomes that estimates before the policy change are statistically insignificant and fluctuate around zero. This evidence suggests that differences in pretreatment trends or anticipatory behavior are not important in my setting. After the policy change, small firms expand their production relative to

²⁵ Supplemental Appendix Table D.1 presents direct effect results for total sales. When I exclude firms with volatile pretreatment export shares, the coefficient estimates imply a positive but imprecisely estimated output response.

²⁶ Supplemental Appendix Table D.2 presents further robustness checks for different sets of controls and for the employment rank and log sales as the firm size measures. The evidence is consistent with Curtis et al. (2021), who do not find differences for a set of smaller firms compared to their main sample of larger firms.

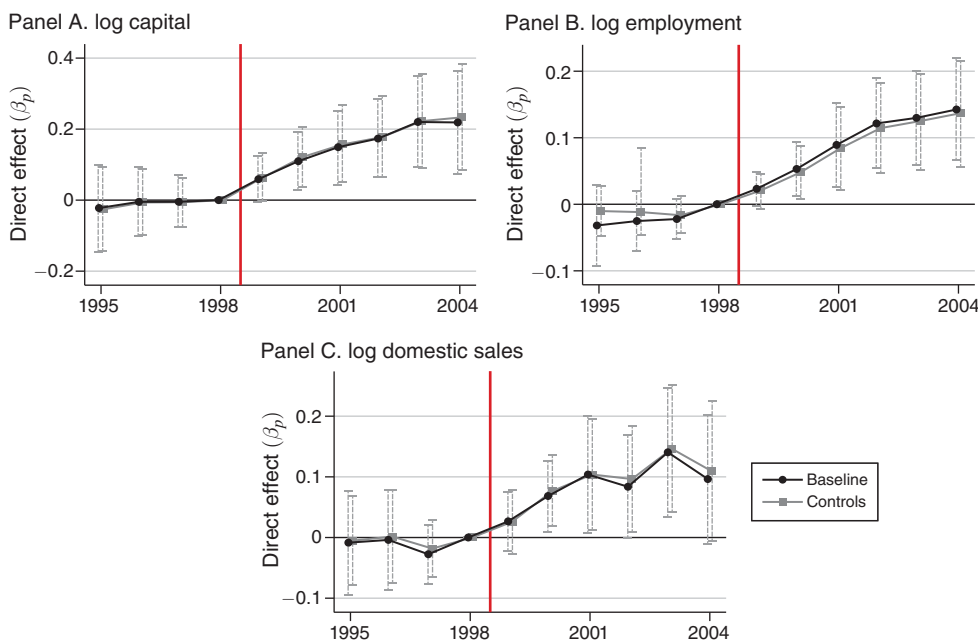


FIGURE 4. DYNAMIC DIRECT EFFECTS OF INVESTMENT TAX CREDITS

Notes: The figure plots yearly direct effect estimates β_p from regressions of specification $Outcome_{it} = \sum_{p=1995}^{2004} \beta_p Small_{i,98} \times \mathbf{1}_{\{t=p\}} + \psi_i + \psi_{nt} + \psi_{st} + \epsilon_{it}$. The coefficient for 1998 is set to zero. The “Control” specification adds pretreatment trends and industry and county characteristics interacted with year fixed effects. Details about the control variables are provided in Table 2. The dependent variables are the log value of the capital stock in panel A, log employment in panel B, and log domestic sales in panel C. Results are based on the AFID data. The dashed lines show the 95 percent confidence interval for each estimate, with standard errors clustered at the county level.

large ones. The adjustments take place over several years, with the effects on inputs and output stabilizing after three to five years.

Throughout the paper, I focus on the capital stock as an input factor to production. Nevertheless, for a comparison of my results with findings in the investment literature, Table 3 provides estimates for the effects of the policy change on investment. For log investment, the baseline estimate in column 1 amounts to an effect of 23 log points. With the user cost of capital decreasing by 15.6 percent for small firms and by 8 percent for large firms (see Section IA), the implied elasticity of investment with respect to the net of tax cost is 3. Results for the investment rate, which takes the extensive margin of investment into account, confirm this positive effect. Given an average investment rate of small firms in the pretreatment period of 0.072, the baseline estimate in column 4 translates to an elasticity of 2. These results remain almost unchanged when including additional controls.

C. Refundability and Worker Types

An unusual feature of the policy is the refundability of investment tax credits. This feature allows even firms without taxable income to immediately obtain a tax deduction (Auerbach 2009). To explore this feature of the policy, I use supplementary information

TABLE 3—INVESTMENT RESPONSE OF FIRMS

	log investment				Investment rate			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small firm $\times$ post	0.230 (0.095)	0.222 (0.092)	0.255 (0.091)	0.247 (0.100)	0.011 (0.004)	0.011 (0.004)	0.012 (0.004)	0.012 (0.005)
Observations	15,590	15,574	15,574	15,574	15,094	15,077	15,077	15,077
Pretreat. trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
$X_n, X_c \times$ year FE	No	No	Yes	No	No	No	Yes	No
County-year FE	No	No	No	Yes	No	No	No	Yes

Notes: The table reports estimates from regressions of specification (1). Details about the control variables are provided in Table 2. The dependent variables are log investment in columns 1–4 and investment over lagged capital winsorized at the ninety-fifth percentile in columns 5–8. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

from the *Kostenstrukturerhebung* dataset, which allows me to split firms by their average net income in the pretreatment period into a positive and negative income group.²⁷ Supplemental Appendix Table D.3 shows that firms in both groups experience positive production effects. For instance, the estimated employment effect is 8.7 log points for the negative income group and 11.7 log points for the positive income group. Furthermore, using pretreatment net income over employment as a continuous measure of firm profitability, I do not find a meaningful heterogeneous response across the whole range of firm profitability. Overall, these results suggest that refundability can be a meaningful feature of investment tax incentives for supporting firms without taxable income.

The literature on skill- and routine-biased technological change highlights the possibility of demand shifts toward skilled labor performing abstract tasks, which complement the use of technology, and away from routine tasks, which machines can easily codify (Katz and Murphy 1992; Acemoglu and Autor 2011). By encouraging the use of capital, the policy may accelerate this substitution of workers.²⁸ To explore this prediction, I rely on the education and occupation information of workers in the BeH data.²⁹

Supplemental Appendix Table D.4 shows that small shifts for college-educated versus non-college-educated workers and abstract-task-intensive versus routine-task-intensive occupations are evident and in the expected direction, but they do not lead to meaningful deviations from the overall employment effect (with the overall effect being comparable to that from the AFiD data). To examine the role of new technology more directly, I use auxiliary data on the information and communication technology (ICT) capital stock for 13 manufacturing industry groups from EU KLEMS (Supplemental Appendix F.9) and divide firms into below- and

²⁷ I determine firms' net income as total sales minus the cost for used intermediate and resold final goods and other costs. As the information is from administrative surveys, the classification of firms, particularly those at the margin, into income groups can deviate from one based on the true taxable income.

²⁸ An extension of the firm production model with two types of labor provides predictions for differential employment responses (see Supplemental Appendix B.5 for details).

²⁹ I classify occupations according to their task content into routine-task-intensive occupations and abstract-task-intensive occupations. This classification is based on Dengler, Matthes, and Paulus (2014), who apply the approach from Autor, Levy, and Murnane (2003) to a representative database with occupational job task requirements for Germany. Since technology could not replace nonroutine manual tasks during the sample period, I group them with abstract tasks.

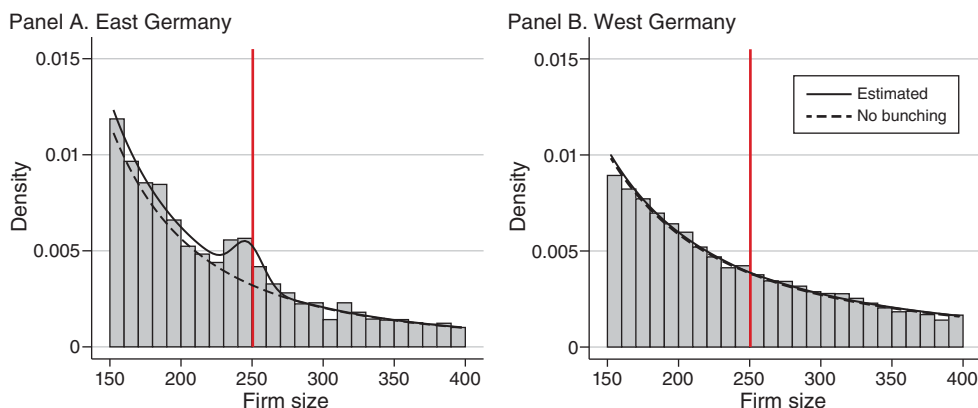


FIGURE 5. FIRM SIZE DISTRIBUTION FOR THE MANUFACTURING SECTOR, 1999–2004

Notes: The figure shows the size distribution of manufacturing firms with 150–400 employees in East Germany (panel A) and West Germany (panel B), respectively. The bars represent the number of firm observations in bins of ten employees for the period 1999–2004, with values normalized to a density function. The solid black line depicts the firm size distribution estimated by maximum likelihood. The dashed black line is the counterfactual firm size distribution without firm bunching and distributional shift. Results are based on the AFiD data.

above-median groups according to the ICT capital share at the industry level in 1998. This split reveals larger differences across worker types in the high ICT reliance group, with the most pronounced shift being toward college-educated workers. Nonetheless, even in this industry group, non-college-educated workers and workers in routine-task-intensive occupations—who stand to lose the most—still benefit from positive demand effects.

D. Evidence from Bunching at the Firm Size Cutoff

So far, the empirical strategy exploits the response of small and large firms away from the policy firm size cutoff. An analysis of the firm behavior right around the cutoff provides a complementary empirical approach, as the jump in the investment tax credit rate incentivizes large firms to bunch at the cutoff for obtaining larger tax deductions.

I explore this behavior by estimating the firm size distribution for the manufacturing sector in the period after the policy change, for which the sample and the difference in the investment tax credit rates across firm size are both larger. To do so, I set up a Lucas (1978) span-of-control model with size-dependent cost of capital (see Supplemental Appendix B.6 for full details). While theory predicts sharp bunching at the cutoff, empirically I follow Garicano, Lelarge, and Van Reenen (2016) by factoring in measurement error that smooths out the observed empirical distribution.

Figure 5 provides a summary, with panel A showing the size distribution of East German manufacturing firms. The bars represent the actual distribution in bins of ten employees, and the solid black line shows the estimated firm size distribution from maximum likelihood.³⁰ Away from the cutoff the density decreases as firm

³⁰ Supplemental Appendix Table D.5 summarizes the underlying parameters of the maximum likelihood estimation for East and West Germany.

size increases. However, this pattern breaks just below the firm size cutoff, where firms bunch roughly between 230 and 250 employees. Evidently, firms react to the jump in the investment tax credit rate by actively choosing a firm size just below the cutoff. The estimated distribution further reveals a rightward shift in the distribution of small firms away from the cutoff (compared to the counterfactual dashed black line), in line with the positive employment effect on small firms found in the difference-in-differences setting.

For comparison, panel B presents the same evidence for West German manufacturing firms. As these firms were generally ineligible for investment tax credits, the firm size distribution should not show any of the previous deviations. A visible inspection and the estimated firm size distribution both confirm that the density is flat around the cutoff and does not have a rightward shift of small firms.

IV. Indirect Effects on Local Manufacturing

Given the positive impact of investment tax credits on directly affected firms, I now turn to the analysis of indirect general equilibrium effects, with a focus in this section on spillovers on local manufacturing.

A long-standing literature identifies agglomeration spillovers as an important factor for local production since they can explain spatially concentrated economic activity despite the associated disadvantages, such as higher costs for labor and land. Marshall (1890) divides cost and productivity advantages from agglomeration into three broad categories: thick labor markets, which reduce search frictions and facilitate better worker-firm matches; knowledge spillovers resulting from intensified interaction between firms or workers; and production advantages from being located close to upstream suppliers and downstream customers (for an overview of theories, see Duranton and Puga 2004). In the context of investment tax incentives, the production expansion of directly affected firms may strengthen these agglomeration advantages, which in turn would generate additional indirect production benefits throughout a region.

Another important factor that shapes the local indirect effect is the response of local labor supply. If local labor supply is not perfectly elastic, due to limited movement of workers across regions or in and out of nonemployment, a change in labor demand increases wages, and thus the production cost, for all firms in a region, independent of their investment tax incentives received.³¹

A. Indirect Effect Specification

To analyze such local spillovers, I extend the previous specification to

$$(2) \quad Outcome_{it} = \beta Small_{i,98} \times Post98_t + \eta ShSmall_{-ir,98} \times Post98_t \\ + \psi_i + \psi_{nt} + \psi_{st} + \epsilon_{it},$$

³¹Mian and Sufi (2014) show that wage and price rigidity can change predictions for employment outcomes. With rigidities, long-run effects after an adjustment of prices would be informative about the local labor supply response.

TABLE 4—LOCAL SPILLOVER EFFECTS OF INVESTMENT TAX CREDITS ON MANUFACTURING FIRMS

	log capital			log employment			log domestic sales	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small firm $\times$ post	0.155 (0.056)	0.147 (0.055)	0.170 (0.054)	0.110 (0.035)	0.095 (0.027)	0.097 (0.028)	0.095 (0.044)	0.098 (0.041)
Share small county $\times$ post	0.247 (0.087)	0.238 (0.086)	0.199 (0.103)	0.103 (0.067)	0.097 (0.042)	0.090 (0.048)	0.060 (0.073)	0.071 (0.079)
Observations	15,146	15,128	15,128	16,514	16,497	16,497	16,482	16,468
Pretreat. trends	No	Yes	Yes	No	Yes	Yes	No	Yes
$X_n, X_c \times$ year FE	No	No	Yes	No	No	Yes	No	Yes

Notes: The table reports estimates from regressions of specification (2). Details about the control variables are provided in Table 2. The dependent variables are log value of capital stock in columns 1–3, log employment in columns 4–6, and log domestic sales in columns 7–8. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

where the new term $ShSmall_{-ir,98}$ is the employment share of small firms in the manufacturing sector in county r in 1998, with the employees of firm i excluded ($\hat{=}$ employment in other manufacturing firms with at most 250 employees divided by employment in all other manufacturing firms in a county). This share of small firms is equal to the (weighted) leave-out mean of $Small_{i,98}$, the variable identifying the direct effect (Huber 2018, 2023).

By interacting the share with the treatment period dummy $Post98_t$, I identify the indirect effects in a difference-in-differences design with a continuous treatment variable. The coefficients of interest are β , the direct effect of higher tax credit rates as in specification (1), and η , which identifies the local spillover effects of investment tax credits. The included fixed effects help control for unrelated shocks that might be correlated with the indirect effects. For example, as industries tend to be spatially clustered (Duranton and Overman 2005; Ellison, Glaeser, and Kerr 2010), industry-year fixed effects can capture industry-specific firm trends such as technology adoption, which otherwise might be attributed to the indirect effect.

B. Main Indirect Effect Results

Table 4 reports results from regressions of specification (2). The first row of estimates capture the direct effects of the investment tax credit policy. For all outcomes, the coefficient estimates are close in magnitude to the analysis before. The second row presents the indirect effect estimates. The estimates are all positive, implying that firms surrounded by a larger share of small firms scale up their production more, independent of their own received investment tax incentives. The effects are robust to controlling for pretreatment trends and including county-specific and industry-specific characteristics interacted with year fixed effects. Including pretreatment trends again reduces standard errors. I therefore focus on this specification in all subsequent average effect analyses.

The indirect effect estimates imply that East German manufacturing firms benefit unevenly from local spillovers. A standard deviation increase in the share of small firms leads to a 3.8 log point ($= 0.238 \times 0.16$) higher capital stock and to 1.6 log points ($= 0.097 \times 0.16$) higher employment for a firm. The estimates

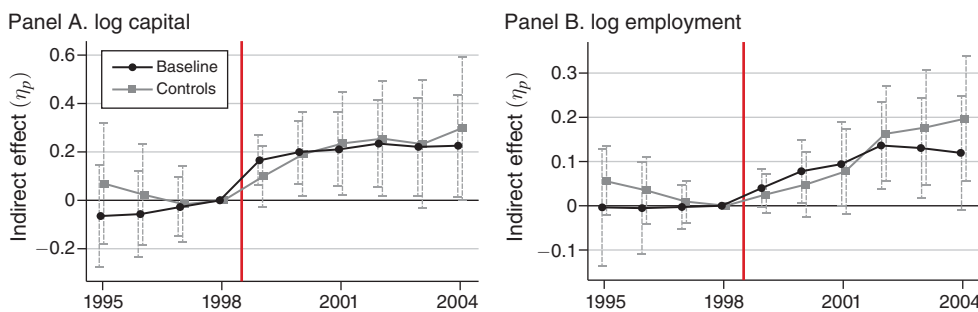


FIGURE 6. DYNAMIC EFFECTS OF LOCAL SPILLOVERS

Notes: The figure plots yearly indirect effect estimates η_p from regressions of specification $Outcome_{it} = \sum_{p=1995}^{2004} \beta_p Small_{i,98} \times \mathbf{1}_{\{t=p\}} + \sum_{p=1995}^{2004} \eta_p ShSmall_{-ir,98} \times \mathbf{1}_{\{t=p\}} + \psi_i + \psi_{it} + \psi_{st} + \epsilon_{it}$. The coefficients for 1998 are set to zero. The “Control” specification adds pretreatment trends and industry and county characteristics interacted with year fixed effects. Details about the control variables are provided in Table 2. The dependent variables are the log value of capital stock in panel A and log employment in panel B. Results are based on the AFiD data. The dashed lines show the 95 percent confidence interval for each estimate, with standard errors clustered at the county level.

can also be related to the policy effect on the average East German manufacturing firm, for which the share of small firms in a county is 66 percent. Because of the larger reduction in the cost of capital for small firms, this firm increases its capital stock by 15.7 log points ($= 0.238 \times 0.66$) and its employment by 6.4 log points ($= 0.097 \times 0.66$) through local spillovers. Both comparisons show that investment tax credits generate sizable additional positive local spillovers for firms and that the direct effects alone understate the impact on local manufacturing. For output, I find a similar indirect firm response, but the effects are less precisely estimated. I also implement the analysis for total sales, for which the indirect effect is positive and even larger in magnitude (see Supplemental Appendix Table D.6).

Figure 6 presents the corresponding yearly indirect effect estimates for capital and employment from an event-study framework. For both outcomes, firms move in parallel before the policy change, with estimates being close to zero and statistically insignificant. Directly after the policy change, the estimates start to increase, with the full effect materializing within two to four years. The dynamic effects on domestic and total sales follow the exact same pattern, but have larger confidence intervals around the coefficient estimates (see Supplemental Appendix Figure D.1).

I corroborate the average effect results in further robustness checks. Supplemental Appendix Figure D.2 summarizes evidence for different sample selection criteria. I find that the range of the included smallest and largest firms has a limited impact on the estimates, consistent with the evidence from the robustness check that controls for heterogeneous effects by firm size. The same finding holds for different ranges of excluded firm sizes around the cutoff. Moreover, the exclusion of multiestablishment firms, which may be active in different counties and industries, does not lead to meaningful changes.³² I also show that including industry-year and federal state-year fixed effects has a material impact only for the sales outcome.

³² Multiestablishment firms with plants in West and East Germany may exploit investment tax credits by shifting production toward East Germany. However, among all firms in the sample, only three percent could do so.

Supplemental Appendix Table D.7 presents results for a specification that interacts the indirect effect with the demeaned local labor share, which the production model shows may spuriously bias the spillover estimates. Supplemental Appendix Table D.8 shows results for a specification with industry-year fixed effects based on a finer classification of 53 (instead of 14) industry groups, which include at least one large firm in every group. Finally, in Supplemental Appendix Table D.9, I explore the impact of different trade exposure variables at the county, industry, and firm level. For all of these additional robustness checks, both the direct and indirect effects remain qualitatively the same.

C. Spillover Effect Channels

Agglomeration economies are a natural candidate for explaining the positive indirect effects on the production of local manufacturing firms. Marshall's proposed agglomeration channels emphasize economic closeness between firms for creating agglomeration benefits. If firms are linked through the same production network, share similar labor inputs, or benefit from similar innovation processes, for example, because they are part of the same broad industry, local spillovers should be stronger. For firms without those connections, spillovers should be more limited even within the same county.³³ Exploring the heterogeneity of the indirect effects therefore is useful to identify the source of the spillovers.

In a first analysis, I divide manufacturing firms into six broad industry groups and decompose the employment share of small firms in the manufacturing sector into

$$(3) \quad ShSmall_{-ir,98} = \frac{EmpSmall_{-inr,98}}{EmpTot_{-ir,98}} + \frac{EmpSmall_{n'r,98}}{EmpTot_{-ir,98}},$$

where employment in surrounding small manufacturing firms is split into a part of the same broad industry $EmpSmall_{-inr,98}$ and a part of other industries $EmpSmall_{n'r,98}$ divided by the total manufacturing employment in surrounding firms $EmpTot_{-ir,98}$.³⁴

When substituting the decomposed share in specification (2) with both employment rates, I can estimate the effect on firms of an increase in the share of small firms originating from small firms in the same industry group (η_1), while holding constant the influence of small firms in the other groups (η_2), and vice versa. With industry affiliation as a proxy for economic closeness, the prediction is that an increase in the share of small firms creates larger indirect effects if the increase originates from firms in the same industry group rather than ones in other industry groups ($\eta_1 > \eta_2$). The specification follows from my production model when I weight firms according to their importance in the generation of spillovers for another firm. The split by industry groups assumes two different weights, a weight for firm pairs in the same broad industry group and one for firm pairs in distinct industry groups (see Supplemental Appendix Section B.7).

³³ Some spillovers, such as improved access to local external finance or labor sharing for administrative and managerial positions, may create positive effects independent of the economic closeness between firms and can explain spillovers to less-connected firms.

³⁴ The broad industry groups are chemical products; metal products; equipment; electronics; food, beverages, and apparel; and other manufacturing.

TABLE 5—LOCAL SPILLOVER EFFECTS WITHIN AND ACROSS INDUSTRIES

	log capital		log employment		log domestic sales	
	(1)	(2)	(3)	(4)	(5)	(6)
Small firm $\times$ post	0.147 (0.054)	0.168 (0.054)	0.095 (0.027)	0.097 (0.028)	0.092 (0.040)	0.097 (0.041)
Share small county $\times$ post						
Part same industry	0.748 (0.249)	0.633 (0.258)	0.172 (0.088)	0.160 (0.091)	0.250 (0.124)	0.247 (0.135)
Part other industries	0.123 (0.101)	0.100 (0.118)	0.080 (0.044)	0.073 (0.051)	0.006 (0.067)	0.030 (0.083)
p -value same = other	0.030	0.071	0.315	0.350	0.070	0.105
Observations	15,128	15,128	16,497	16,497	16,468	16,468
Pretreat. trends	Yes	Yes	Yes	Yes	Yes	Yes
$X_{it}X_c \times$ year FE	No	Yes	No	Yes	No	Yes

Notes: The table reports estimates from regressions of specification (2), with the share of small firms split according to equation (3) into a part of small firms in the same industry group and a part of small firms in other industry groups. Industries are grouped into chemical products; metal products; equipment; electronics; food, beverages, and apparel; and other manufacturing. Details about the control variables are provided in Table 2. The row p -value same = other reports the p -values for a test of the equality between the same industry and other industry coefficients. The dependent variables are log value of capital stock in columns 1–2, log employment in columns 3–4, and log domestic sales in columns 5–6. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

Table 5 shows that for all outcome variables the indirect effect estimates for small firms in the same industry group are larger than the estimates for those in other industries. This difference is particularly large for capital, with coefficient estimates of 0.748 for the same industry group and 0.123 for the other industry group (column 1), and for domestic sales with estimates of 0.250 and 0.006 (column 5), respectively. For employment, the effect from an increase in the rate of small firms in the same industry group is still about twice as large as such a change in another industry. Thus, a one percentage point increase in the employment rate of small firms in the same industry group would generate twice the indirect effect from the policy change than that increase in another industry group. Overall, the results provide evidence that the economic closeness between firms is critical for the transmission of local spillovers (for the dynamic effects, see Supplemental Appendix Figure D.3).³⁵ In addition, the heterogeneous response provides evidence against mechanical bias, which would generate equal effects for all industry groups (Huber 2023).

In a second analysis, for evidence on the specific channels of the local spillovers, I use weighted shares of small firms

$$(4) \quad ShSmall_{-inr}^M = \sum_{n'} w_{n,n'}^M \frac{EmpSmall_{-in'r}}{TotEmp_{-ir}},$$

³⁵ While firms in the same narrow industry are more likely product-market rivals, when I assign firms in the same four-digit industry to a separate third group, results for the same broad industry group and the other industry groups do not substantially change (see Supplemental Appendix Table D.10). I also obtain the same qualitative results for a specification with employment shares of small firms in a given industry and county (instead of the rates) and the relative size of an industry in a county interacted with the treatment period dummy (see Supplemental Appendix Table D.11).

where the weight $w_{n,n'}^M$ between industries n and n' is given by different industry closeness measures. This specification corresponds to a version of my firm production model that allows for differences in the generation of spillovers across industries according to the closeness measures. Specifically, the weighted shares imply that the difference in the indirect effects for one additional small manufacturing firm is proportional to the closeness measure of this firm.

For the role of local input-output linkages, I construct two closeness measures from the German input-output tables for the 31 available manufacturing industry groups for the pretreatment period (Supplemental Appendix F.10). To capture the closeness to downstream customers in production, I use the share of total output of industry n sold to industry n' . For the closeness to upstream suppliers, I take the share of total inputs of industry n bought from industry n' . The higher the dependence of industry n on the sales to industry n' or on the intermediate goods as input from industry n' , the larger should the local spillovers from firms in industry n' be.

As a third measure, I determine job-to-job movers across three-digit industries from the BeH employer-employee data for the pretreatment period. I take the average of the share of workers from industry n moving to industry n' , and the share of workers from industry n' moving to industry n . I define a job mover as a person employed in different establishments in consecutive years. This measure is related to both joint labor sharing and knowledge spillovers. Observing a higher share of workers moving between industries suggests closeness in the sourcing of labor inputs. At the same time, by transmitting firm-specific knowledge from one employer to the next, workers also enable knowledge spillovers across industries.

As the construction of the three closeness measures leads to differences in their scale, to allow for a comparison of estimates, I rescale the closeness measures to a standard deviation of one. Table 6 presents results from specifications with all three closeness measures included jointly. Coefficient estimates are consistently positive only for the downstream customer channel, and these estimates are also always larger than those for the two other closeness measures. Thus, an increase in the share of small firms implies particularly large additional spillover effects when the increase is from small firms in downstream industries. One explanation for the asymmetry in the effects on the production network is that the scaling up of production is closest to a demand shock, which propagates primarily to intermediate input suppliers (Acemoglu, Akcigit, and Kerr 2016). In contrast, labor sharing cannot explain the indirect effects. Knowledge spillovers may be more relevant in settings that create shocks to firm productivity. Moreover, given high unemployment in East Germany, if new workers are hired locally from unemployment, the local labor pool does not change.³⁶

Supplemental Appendix Table D.13 provides supporting evidence for the importance of local downstream customers. I implement a version of the indirect effect specification that allows for a heterogeneous response of nonexporting and exporting firms. If the indirect effects are driven by local downstream customers, then

³⁶ When I include each closeness measure separately, local input-output linkages remain the channel that creates the strongest effects. The results are robust to including the unweighted share of small firms. Both input sharing of intermediate goods and capital goods as alternative channels cannot explain the indirect effects, and an alternative closeness measure for labor sharing based on the similarity of industries' occupation structure leaves results qualitatively unchanged (see Supplemental Appendix Table D.12).

TABLE 6—LOCAL SPILLOVERS BY MEASURES OF INDUSTRY CLOSENESS

	log capital		log employment		log domestic sales	
	(1)	(2)	(3)	(4)	(5)	(6)
Small firm $\times$ post	0.149 (0.055)	0.170 (0.055)	0.095 (0.028)	0.093 (0.029)	0.082 (0.041)	0.086 (0.041)
Share small county (weighted) $\times$ post						
Downstream customer	0.131 (0.120)	0.114 (0.125)	0.079 (0.047)	0.102 (0.047)	0.199 (0.083)	0.241 (0.088)
Upstream supplier	0.015 (0.135)	0.013 (0.133)	−0.038 (0.049)	−0.066 (0.049)	−0.082 (0.084)	−0.132 (0.087)
Labor sharing	−0.010 (0.067)	0.003 (0.062)	−0.025 (0.019)	−0.024 (0.022)	−0.084 (0.037)	−0.075 (0.039)
p -value down = up	0.634	0.684	0.205	0.067	0.083	0.028
p -value down = sharing	0.301	0.450	0.050	0.024	0.004	0.002
Observations	15,128	15,128	16,497	16,497	16,468	16,468
Pretreat. trends	Yes	Yes	Yes	Yes	Yes	Yes
$X_{nt}X_c \times$ year FE	No	Yes	No	Yes	No	Yes

Notes: The table reports estimates from regressions of specification (2), with small firms in the calculation of the share of small firms weighted by different industry closeness measures according to equation (4). The industry closeness measures are the share of total output of industry n sold to industry n' (downstream customer), the share of total inputs of industry n bought from industry n' (upstream supplier), and the average of the share of workers from industry n moving to industry n' and the share of workers from industry n' moving to industry n (labor sharing). Details about the control variables are provided in Table 2. p -value down = up and p -value down = sharing report the p -values for a test of equality between the downstream customer and upstream supplier coefficients and the downstream customer and labor sharing coefficients, respectively. The dependent variables are log value of capital stock in columns 1–2, log employment in columns 3–4, and log domestic sales in columns 5–6. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

exporting firms, which are less dependent on selling output locally, should benefit less. The results confirm this prediction, with the indirect effects on nonexporting firms being substantially larger than on exporting firms.

The results point to agglomeration economies as an explanation for the positive spillover effects on local manufacturing firms. At the same time, frictions in the local labor supply may dampen the overall impact. An indication for the relevance of this channel is the materially larger effect of the spillovers on capital than on employment. This difference may be explained by imperfectly elastic local labor supply, which would increase local wages and incentivize all firms in a county to shift production away from labor toward capital. To further explore this channel, I estimate the effect of the policy change on the average wages paid by firms. Supplemental Appendix Table D.14 shows that the indirect wage effect is positive and becomes statistically significant with the inclusion of county and industry characteristics interacted with year fixed effects. The policy change therefore creates an opposing effect on production cost through the increase in local manufacturing wages. However, this effect is not strong enough to offset the positive spillovers in my setting.³⁷

³⁷Estimates from a dynamic specification suggest a limited role of wage rigidity in my setting. Previous studies commonly focus on downward rigidity, which is relevant in settings with negative local shocks (Altonji and Devereux 1999). The wage analysis also reveals that small firms reduce their wage relative to large ones. In an analysis of log wage residuals from the BeH employer-employee data, which allows me to control for different sets of observable worker characteristics, I find that this negative wage effect is primarily driven by an increase in the number of employees with lower firm tenure.

TABLE 7—LOCAL SPILLOVERS BY DISTANCE

	log employment							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small firm $\times$ post	0.120 (0.023)	0.119 (0.025)	0.127 (0.024)	0.128 (0.024)	0.124 (0.024)	0.124 (0.024)	0.122 (0.025)	0.122 (0.026)
Share small $\times$ post								
Within county	0.095 (0.040)							
Within (0 km, 2 km)		0.052 (0.024)					0.057 (0.025)	0.050 (0.026)
Within [2 km, 5 km)			0.059 (0.022)				0.063 (0.025)	0.051 (0.026)
Within [5 km, 10 km)				−0.014 (0.026)			−0.010 (0.027)	−0.025 (0.030)
Within [10 km, 25 km)					−0.013 (0.053)		−0.039 (0.056)	−0.058 (0.058)
Within [25 km, 50 km)						0.023 (0.082)	0.026 (0.084)	−0.002 (0.086)
Observations	20,590	18,760	19,120	20,350	20,590	20,590	17,520	17,450
Pre.-treat. trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$X_{it}X_c \times$ year FE	No	No	No	No	No	No	No	Yes

Notes: The table reports estimates from regressions of specification (2), with the employment share of small firms in the manufacturing sector either calculated at the county level or for different distance intervals. Details about the control variables are provided in Table 2. The dependent variable is log employment throughout. Results are based on the BeH data. Standard errors in parentheses are clustered at the county level.

D. Geographic Scope of Local Spillovers

Another important question for the understanding of agglomeration economies is how quickly the spillover effects attenuate across space. While the county-level analysis distinguishes relatively small regions, spillovers may operate at even smaller scales and arise for firms across administrative boundaries. To investigate the spatial propagation, I apply the estimation strategy to flexible distance intervals around each firm. The calculation of these intervals builds on geolocation data (IEB GEO) from the Institute for Employment Research (Supplemental Appendix F.11) Starting in 1999, the dataset includes establishments' geographic coordinates generated from addresses available in the German Federal Employment Agency registries. Additionally, I rely on municipality and postal code data (Supplemental Appendix F.12 and F.13) to infer establishment locations for 1998 (see Supplemental Appendix C.4). I determine the shortest (great-circle) distance between establishments and calculate the employment share of small firms in the manufacturing sector for different distance intervals. As I can calculate these measures only for the BeH data, I use employment as the firm outcome.

Table 7 presents estimation results for this analysis. I confirm in column 1 that the spillover effects at the county level are comparable to the results obtained from the AFiD data. Columns 2–6 show the results, one by one, for different distance intervals around each firm. For a distance interval of up to 2 kilometers (approximately 1.2 miles), the estimate is 0.052. The magnitude amounts to more than half of the county level effect, pointing to the importance of nearby firms for generating the local spillover effects. For distances of 2–5 kilometers (approximately 1.2–3.1 miles), the estimate is 0.059. In contrast, for greater distances, the estimates

are small and statistically insignificant. When I include all distance measures in the same specification to take into account any dependence between the different distance intervals, and when I further include additional control variables, the estimates remain largely unchanged (columns 7–8).

The results suggest a strong gradient of local spillovers with the predominant impact originating from nearby firms. Since the indirect effects are primarily generated from local input-output linkages, the analysis is informative about the geography of production networks. The organization of production in East Germany during the time of separation, with geographical clusters of connected firms being centrally managed as *Kombinate*, may have had a lasting effect on the production network of firms even after reunification.

E. Placebo Analysis and Alternative Support Program

With the investment tax credit policy targeting firms in East Germany, applying the estimation strategy to West German manufacturing firms in a placebo analysis can provide insights about the probable firm response without the policy change. However, as the descriptive evidence has shown, manufacturing firms in both parts of Germany followed distinct trends, making such an analysis less representative for East German firms. A closer inspection of production trends in West Germany further shows important differences within the manufacturing sector.³⁸ I therefore provide placebo results for a set of comparable manufacturing firms. Conditional on the main sample restrictions, I match each East German firm to the West German firm with the lowest Mahalanobis distance in terms of their pretreatment period employment and average wage, yearly employment and wage growth, and the average firm size of the labor market. Compared to all West German firms in the matching process, the selected firms are smaller and have higher employment growth during the pretreatment period.

Columns 1–3 of Table 8 summarize the results for the sample of matched West German firms. Coefficient estimates for all outcomes are statistically insignificant and close to zero, consistent with these firms being ineligible for investment tax credits. I can even reject at conventional levels that the direct effects are equal to those for East German manufacturing firms. In support of these results, columns 4–6 show that estimates are substantially smaller than those for the main sample in a placebo analysis of manufacturing firms located in West German counties close to the former inner German border, mapped in Supplemental Appendix Figure C.1. This sample consists of firms for which the German division and reunification had a particularly strong impact on market access (Redding and Sturm 2008). Furthermore, for a dynamic specification, coefficient estimates for both sets of firms do not show any break in trends around the time of the policy change (see Supplemental Appendix Figure D.4). Thus, without the policy change, East German manufacturing firms would likely not have experienced the positive production effects.

³⁸The direct effects for the complete sample of West German manufacturing firms are close to zero when I exclude firms with extreme employment changes and control for heterogeneous effects by firm size (see Supplemental Appendix Table D.15).

TABLE 8—PLACEBO ANALYSIS OF WEST GERMAN MANUFACTURING FIRMS

	Matched firms			Firms close to inner German border		
	log capital (1)	log employment (2)	log domestic sales (3)	log capital (4)	log employment (5)	log domestic sales (6)
Small firm $\times$ post	−0.001 (0.040)	0.017 (0.022)	−0.007 (0.034)	−0.032 (0.039)	0.040 (0.018)	0.008 (0.028)
<i>p</i> -value West = East	[0.030]	[0.024]	[0.061]	[0.009]	[0.091]	[0.090]
Share small county $\times$ post	0.017 (0.143)	0.023 (0.051)	0.028 (0.070)	0.130 (0.111)	−0.004 (0.056)	0.009 (0.086)
<i>p</i> -value West = East	[0.185]	[0.262]	[0.807]	[0.442]	[0.148]	[0.694]
Observations	16,184	16,498	16,471	13,504	15,976	15,955
Pre-treat. trends	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates from regressions of specification (2) for West German manufacturing firms. Details about the control variables are provided in Table 2. The dependent variables are log value of capital stock in columns 1–4, log employment in columns 2–5, and log domestic sales in columns 3–6. The *p*-values are for a test of equality of the coefficients for West and East German firms in a fully interacted model including firms of the main sample. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

In an additional analysis, I explore the impact of an alternative support program. Government support for East Germany was widespread after German reunification, with programs targeting, for example, infrastructure and housing projects. For firms, the *Gemeinschaftsaufgabe Verbesserung der regionalen Wirtschaftsstruktur* (GRW) provided investment subsidies at the discretion of state governments. During the sample period, changes to the maximum subsidy rates occurred for a subset of East German counties in 1997 and 2000 (Siegloch, Wehrhöfer, and Etzel 2025), potentially confounding my identification strategy. From official government reports, I build an indicator for all counties and years affected by the decrease in GRW subsidy rates and include it and its interaction with a small firm dummy in the main estimation specification.

For this specification, I find no material impact on the effects of the investment tax credit policy (see Supplemental Appendix Table D.16). Thus, the GRW program likely does not confound the analysis of investment tax credits. For the changes in the maximum subsidy rates of the GRW program, I do not find any strong effect on firm production. This finding may be attributable to the program prioritizing large-scale investment projects and plant openings, thus targeting firms different from those in the main sample.

V. Indirect Effects beyond Local Manufacturing

A. Local Demand Effects and Nonmanufacturing Employment

The manufacturing sector is only one part of a local economy. With the scaling up of manufacturing production, increased employment may translate to higher aggregate household income in a region. Part of this income will be spent on the consumption of locally provided services, such as restaurants, retail, or personal services, and may in turn lead to the hiring of additional service workers (Moretti 2010).

To explore this spillover channel, I adapt the estimation strategy to the nonmanufacturing sector. Because the investment tax credit rate changes did not apply to

TABLE 9—LOCAL SPILLOVER EFFECTS ON EMPLOYMENT OF NONMANUFACTURING INDUSTRIES

	Firm						County-industry	
	Nonmanufacturing		Local consumer services		Business services		Nonmanufacturing	Manufacturing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Share small county $\times$ post	−0.003 (0.015)	0.006 (0.017)	0.058 (0.028)	0.067 (0.033)	−0.095 (0.048)	−0.086 (0.073)	−0.005 (0.040)	0.209 (0.088)
Observations	129,680	129,110	19,170	19,050	10,030	9,970	55,940	26,680
Share manuf. $\times$ post	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pretreat. trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$X_{nt}X_c \times$ year FE	No	Yes	No	Yes	No	Yes	Yes	Yes

Notes: The table reports estimates from regressions of specification (2), with the regressor for the direct effect being excluded. All regressions add the employment share of the manufacturing sector in a county in 1998 interacted with the treatment period dummy as control. Details about the other control variables are provided in Table 2. Columns 1–6 report results from firm-level data for firms active throughout the sample period. Columns 7–8 report results from county-by-three-digit industry data aggregated from the establishment level, with the sample restricted to county-industry groups that have positive employment throughout the sample period, county-industry fixed effects instead of firm fixed effects, and the observations weighted by employment in 1998. The dependent variable is log employment throughout. The sample includes firms in all nonmanufacturing industries in columns 1, 2, and 7; firms in the retail, restaurant, real estate, and other services activities industries in columns 3–4; firms in the business service industries in columns 5–6; and firms in all manufacturing industries in column 8. Results are based on the BeH data. Standard errors in parentheses are clustered at the county level.

this sector, I estimate only the indirect effects. Because the aggregate impact of the manufacturing sector in a given county depends not only on its share of small firms but also its overall size, I include the manufacturing employment share in 1998 at the county level interacted with the treatment period dummy as another baseline control. With only the BeH data including information on the nonmanufacturing sector, I rely on employment as the outcome variable.

Columns 1–2 of Table 9 report coefficient estimates for the complete sample of nonmanufacturing firms. The estimates imply a negligible overall employment effect. However, this result masks important heterogeneity across industries. Following Mian and Sufi (2014), I classify firms in the retail and restaurant industry as dependent on local consumer demand and add real estate services, which depend on local housing demand, and other service activities, such as hair salons. In line with the predictions for local consumer demand spillovers, the employment effect is positive for these firms (columns 3–4). For the average firm in this sample (for this firm, 70 percent of local manufacturing employment is in small firms), the employment increases by 4.0–4.6 log points.

In contrast, other industries experience negative effects. In columns 5–6, I report results for business services such as IT and R&D services. I find that employment decreases for the average business service firm by 5.7–6.2 log points (for this firm, 65 percent of local manufacturing employment is in small firms). Firms in these industries may face stronger competition from manufacturing firms that start to assume some of these services themselves. While both subgroups constitute a limited share of total nonmanufacturing employment, the results indicate that both positive and negative effects shape the overall effect on the nonmanufacturing sector.

For a comparison of the effects with those on the manufacturing sector, I explore the strength of the effects at a higher level of aggregation. I aggregate the establishment level data at the county-industry level and implement the analogous estimation strategy. The results confirm the differences in effects across sectors found at the firm

level. Column 7 shows that the effect on the nonmanufacturing sector remains close to zero in the aggregate effect specification. In contrast, column 8 highlights the positive employment effects found for the manufacturing sector, with the aggregate effect size corresponding well to the sum of both the direct and indirect effects at the firm level.

B. *Product-Market Spillovers*

A concern with investment tax credits targeted to specific firms is that they create an imbalance across firms. Firms receiving incentives can produce more cost effectively, allowing them to offer lower prices and capture a higher market share. For tradable goods, which can be sold nationwide, even faraway rival firms may lose out through this product-market channel. Given that firms in the same narrow industry are likely to offer the same or at least highly substitutable goods, these spillovers should affect such firms the most.

Similar to the previous specifications, to test for this spillover channel, I implement a difference-in-differences design with the employment share of small firms in the manufacturing sector in 1998 interacted with the treatment period dummy as main regressor. However, this time I determine this share at the four-digit industry level. To capture the specific influence of the policy change, I calculate this share for firms in East Germany. To capture the dependence of industries on East German firms more generally, I include the employment share of East German firms in a given industry in 1998 interacted with the treatment period dummy as control.

Table 10 presents different versions of the described regression specification. Columns 1–2 show estimates for a combination of the local and industry-level spillover analysis for East German manufacturing firms. The coefficient estimates for the direct and local indirect effects remain almost unchanged. Omitting the industry-level spillovers thus does not bias my previous results (Huber 2023). For the product-market channel, the coefficient estimates imply a negative employment effect on manufacturing firms in East Germany. However, compared to the local spillovers, they are considerably weaker. For West German manufacturing firms, estimates for the industry spillovers are similar in magnitude (columns 3–4). Thus, even though the policy targeted only East German firms, it affected firms throughout Germany through within-industry spillovers.

When I separately estimate the effects for West German firms below and above the 1998 sales median, I find that the average effect is driven by the smaller firms and that larger firms are almost completely shielded from these spillovers (column 5). Smaller firms may be less capable of responding to the additional product-market competition created by the policy. While this group is significant in the number of firms, it represents a relatively small share of the overall production in the economy. Consistent with this, when I implement the estimation strategy for aggregate data at the county-by-three-digit industry level, at which larger firms have a higher weight, I find that the effect is close to zero (column 6). Overall, the evidence implies a limited impact of product-market spillovers on the aggregate West German manufacturing sector.

Figure 7 summarizes the dynamic effects of both the local nonmanufacturing spillovers (panel A) and the within-industry spillovers on West German manufacturing firms (panel B). Before the policy change, the coefficient estimates for the

TABLE 10—EFFECTS OF WITHIN-INDUSTRY SPILLOVERS

	Firm					County-industry
	East		West			West
	(1)	(2)	(3)	(4)	(5)	(6)
Small firm $\times$ post	0.098 (0.028)	0.099 (0.029)				
Share small county $\times$ post	0.101 (0.043)	0.085 (0.048)				
Share small industry $\times$ post	−0.030 (0.032)	−0.026 (0.034)	−0.022 (0.011)	−0.023 (0.011)		0.005 (0.043)
Share small industry $\times$ post $\times$ small					−0.045 (0.016)	
Share small industry $\times$ post $\times$ large					−0.002 (0.014)	
Observations	16,397	16,397	135,543	135,543	135,543	100,060
Share East $\times$ post	Yes	Yes	Yes	Yes	Yes	Yes
Pretreat. trends	Yes	Yes	Yes	Yes	Yes	Yes
$X_n, X_c \times$ year FE	No	Yes	No	Yes	Yes	Yes

Notes: The table reports estimates from regressions of a specification that interacts the employment share of small firms in East Germany in a given four-digit industry in 1998 with the treatment period dummy. All regressions control for the employment share of East German firms in a given four-digit industry in 1998 interacted with the treatment period dummy. The specifications in columns 1–2 further integrate the terms of specification (2). All specifications include group (firm or county-industry), industry-year, and federal state-year fixed effects. Details about the other control variables are provided in Table 2. The dependent variable is log employment throughout. Included firms are those in the main sample for East Germany in columns 1–2 and an analogous set of West German firms in columns 3–5. In column 5, separate coefficients are estimated for firms below and above the 1998 sales median. Column 6 reports results for county-by-three-digit industry data aggregated from the establishment-level, with the sample restricted to county-industry groups that have positive employment throughout the sample period, county-industry fixed effects instead of firm fixed effects, the employment shares determined at the three-digit level, and observations weighted by 1998 employment. Results are based on the AFiD data. Standard errors in parentheses are clustered at the county level.

different firm groups are close to zero and statistically insignificant. After the policy change, the estimates start to diverge, with positive and negative employment effects arising in line with the evidence from the average effects. The dynamics show that the effects develop over several years, similar to the dynamics of the direct and indirect effects in the East German manufacturing sector.

C. Worker Flows

The results for the investigated spillover channels suggest a limited overall impact of the policy on firms outside the East German manufacturing sector. Which workers then are moving to East German manufacturing firms to create the positive employment effect? As the aggregate labor pool is fixed, an analysis of worker flows provides a convenient way for exploring this question.

Using the BeH data, I investigate flows of workers across firms by analyzing their status one year before entry and one year after exit from a firm. In a counterfactual exercise, I accumulate yearly employment growth of firms from different worker flow groups according to

$$(5) \quad \ln L_{it}^G = \ln \left[\prod_{p=1996}^t \left(1 + \frac{\Delta L_{ip}^G}{L_{ip-1}} \right) \right],$$

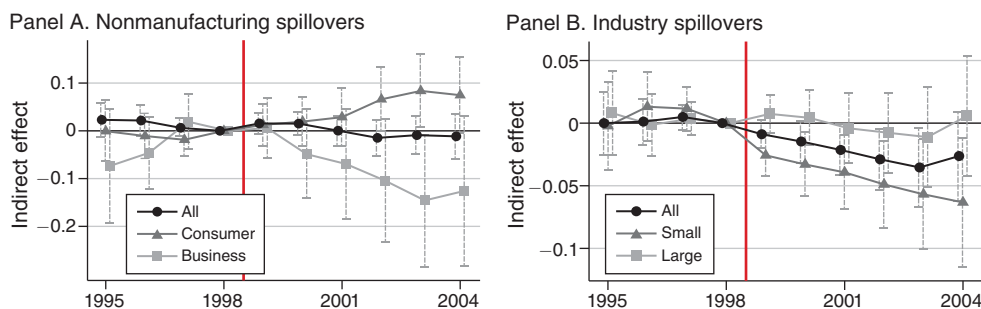


FIGURE 7. DYNAMIC EFFECTS OF LOCAL NONMANUFACTURING AND WITHIN-INDUSTRY SPILLOVERS

Notes: Panel A plots yearly estimates for the employment share of small firms in the manufacturing sector in a given county in 1998 interacted with year fixed effects for a specification with the employment share of the manufacturing sector in a given county in 1998 interacted with year fixed effects as control. The sample consists of firms active throughout the sample period in all nonmanufacturing industries (all), local consumer demand dependent services (consumer), or business services (business). Results are based on the BeH data. Panel B plots yearly estimates for the employment share of small firms in East Germany in a given four-digit industry in 1998 interacted with year fixed effects for a specification with the employment share of East Germany in a given four-digit industry in 1998 interacted with year fixed effects as control. The sample consists of West German manufacturing firms with the analogous restrictions of the main sample. Coefficient estimates are reported for the complete sample (all) and split for firms below (small) and above (large) the 1998 sales median. Results are based on the AFiD data. In both panels, the dependent variable is log employment. Additional controls are firm, industry-year, and federal state-year fixed effects. The coefficients for 1998 are set to zero. The dashed lines show the 95 percent confidence interval for each estimate, with standard errors clustered at the county level.

where ΔL_{ip}^G is the change in the number of employees of subgroup G for year p , and L_{ip-1} is actual firm employment one year before. For overall worker flows, this expression is equivalent to the log of total employment in the estimation, as firm fixed effects subsume the difference in initial employment in 1995.

Supplemental Appendix Table D.17 shows that transitions to and from nonemployment are economically important, explaining 41 percent of the direct and 85 percent of the indirect employment effect. Thus, the policy seems to have helped workers who would have stayed nonemployed or exited to nonemployment. Job-to-job transitions remain important for the direct effect, with worker flows to and from the service sector constituting a sizable share of the direct effect. Interestingly, part of this effect is explained by flows with business services, which see their employment drop through local spillovers. In terms of other adjustment margins, flows to and from West German firms of all sectors are not a significant factor in explaining the effect, nor do I find evidence that shifts between large and small East German manufacturing firms or across manufacturing firms in different East German counties drive the direct and indirect employment effects.

VI. Discussion

The analysis produces a variety of findings that connect to different strands of the economic literature. In this section, I quantitatively and qualitatively compare both the direct and indirect effects with those from previous studies. As a summary of this discussion, I determine the cost per job of the policy.

A. Comparison of Direct Effects

Studies of investment tax incentives commonly determine the elasticity of investment with respect to the net of tax cost. For US bonus depreciation, House and Shapiro (2008) find elasticities between 6 and 14 in a comparison of aggregate investment by capital types.³⁹ Ohrn (2019) estimates the elasticity at 9.6 for the manufacturing sector. Using firm-level data, Zwick and Mahon (2017) calculate an elasticity of 7.2.⁴⁰ For accelerated depreciation allowances in the United Kingdom, Maffini, Xing, and Devereux (2019) find an elasticity of 10.7.

My analysis establishes a substantially lower elasticity. Since my classification of firms by their policy-relevant firm size in 1998 does not account for changes in a firm's investment cost when crossing the firm size cutoff, my estimated investment response might be too low. Following Jakobsen et al. (2020), I therefore estimate the treatment-on-the-treated effect by instrumenting the firm size status in each year with the status in 1998 (see Supplemental Appendix D.18). The implied elasticities increase slightly to 2.4 (based on the investment rate) and 3.5 (based on log investment). The elasticities may also reflect a lower take-up of investment tax credits. However, for US bonus depreciation, Kitchen and Knittel (2016) report a take-up rate for corporations of 50–60 percent. From German corporate tax records for 2004, I find that 70 percent of East German manufacturing firms with a comparable firm size to my main sample claimed investment tax credits.

A remaining difference lies in the design of the specific tax policies. While investment tax credits offer a direct tax deduction of a share of the investment cost, accelerated depreciation allowances achieve this effect indirectly through a shift of deductions from future periods to the present. The larger investment response generated from the shift of deductions may be due to high effective discount rates due to, for example, financial constraints. In line with this reasoning, for a VAT reform in China that allowed firms to deduct fixed investment from the tax base similar to investment tax credits, Chen et al. (2023) find an elasticity of investment of 2.4.

My analysis also allows me to relate the change in the cost of capital to the effects on other firm outcomes. For the treatment-on-the-treated effects, I estimate a capital demand elasticity of $0.181/0.076 = 2.4$ and a labor demand elasticity of $0.116/0.076 = 1.5$ (the user cost of capital decreased by 15.6 percent for small firms and by 8.0 percent for large firms, see Section IA). The qualitative response is similar to findings for US bonus depreciation. Curtis et al. (2021) find that firms scale up production by increasing their capital, overall employment, and employment of different worker types, but they do not experience changes to their productivity. As in their setting, my effects are not immediate but accumulate over time.

³⁹House and Shapiro (2008) determine the investment supply elasticity. In a setting with temporary tax incentives and long-lived capital goods, they show that this elasticity corresponds to the overall investment response.

⁴⁰For very large firms, Zwick and Mahon (2017) determine an elasticity of investment of 4.5 (author's calculation from a semielasticity of investment of 2.27).

B. *Comparison of Indirect Effects*

For modeling agglomeration economies, understanding the underlying mechanisms is crucial. Ellison, Glaeser, and Kerr (2010) identify local input-output linkages as particularly important in an analysis of industry coagglomeration patterns. In contrast, Greenstone, Hornbeck, and Moretti (2010) and Helm (2020) explore firm openings and trade shocks, respectively, and find evidence for the importance of knowledge spillovers and labor sharing but not for local input-output linkages. Similarly, Gathmann, Helm, and Schönberg (2020) find no evidence for local spillovers from input-output linkages after mass layoffs by large firms. One explanation for the different findings is that the variation used in each study for identifying the spillover effects comes from distinct economic shocks. In my setting, the scaling up of production without productivity gains and limited movement of workers across regions is consistent with local input-output linkages being the main source for the local spillovers.

The magnitude of the local spillovers is comparable to those found in previous studies. I determine a local job multiplier of about 2.0 (one indirect job for every directly created one), well within the range from 1.5 to 2.5 of typical values found in the economic literature (see Bartik and Sotherland 2019).⁴¹

C. *Cost per Manufacturing Job*

For a final assessment of the magnitude of the effects, I compare the fiscal cost of the policy to the number of jobs created in East German manufacturing firms. I determine the total number of new jobs by applying the average direct and indirect estimates to aggregate manufacturing employment in East Germany (550 thousand, from Table 1). With small firms employing 66 percent of the workers and receiving close to twice the reduction in the cost of capital of large firms, the policy change translates to 138 thousand new jobs created by small firms and 52 thousand new jobs created by large firms (based on estimates from column 5 of Table 4).

For the calculation of the fiscal cost, I consider the direct tax expenditures from the investment tax credits and the additional depreciation deductions from the non-reduced depreciation base. Information on the latter value are not available from government reports. I therefore calculate the fiscal cost by applying the investment tax credit rate changes to aggregate equipment investment.⁴² I estimate that the policy change led to an increase in the annual cost for the investment tax credits of €449 million, close in magnitude to the change in claimed investment tax credits determined from corporate tax records. The depreciation deductions imply an additional increase in the annual cost of €148 million.

⁴¹ Given the average employment share of small firms in the manufacturing sector, 66 percent of firms (employment weighted) increase employment by 9.5 log points (direct effect), and all firms increase employment by 0.66×9.7 log points (indirect effect).

⁴² Aggregate equipment investment was on average €6.7 billion per year in the pretreatment period and €7.0 billion per year in the treatment period. I determine the cost for a take-up rate of 70 percent, an investment share of small firms of 66 percent, a present value of one dollar of depreciation deduction of 0.88, and an overall tax rate of 50 percent until 2000 and 40 percent thereafter.

I obtain an estimate for the annual cost per manufacturing job of $597/0.190 = \text{€}3,150$, or \$4,900 at 2010 prices. The estimate suggests a modest cost of the investment tax credit policy, particularly in light of the positive effects on nonemployed workers. The local spillovers on the manufacturing sector partially explain the low estimate. If I determine the created jobs only from the direct effect, the cost per job amounts to \$9,800. Although the calculations extrapolate from the specific sample to the entire manufacturing sector and are specific to the institutional setting, I consider the estimate a valuable data point for assessing the magnitude of the employment effect. The cost-per-job estimate is comparable to those from similar policies. Garrett, Ohrn, and Suárez Serrato (2020) determine a total cost per job of \$20,000 to \$50,000 over a ten-year horizon for US bonus depreciation, and Duan and Moon (2024) determine an annual estimate of roughly \$3,000 for Canadian accelerated depreciation allowances. Criscuolo et al. (2019) find an annual cost per job of \$3,500 for a UK investment subsidy program targeting firms in disadvantaged regions.⁴³

VII. Conclusion

In this paper, I investigate the direct and indirect effects of investment tax incentives on firm production. I analyze an investment tax credit policy that was introduced after German reunification in former East Germany. A differential change in the tax credit rates for manufacturing firms by firm size allows me to estimate causal effects. Using a difference-in-differences research design, I estimate the direct effects by comparing the response of small firms to the policy change with that of large firms. I estimate additional indirect spillover effects by comparing the response of firms by the employment share of small firms in the manufacturing sector in a given county or industry.

Overall, my findings provide a favorable assessment of investment tax incentives. Many of the intended effects, such as higher labor demand and positive spillover effects are realized. A positive labor demand response for low-skilled workers, net inflows of workers from nonemployment, and a limited impact on product-market rivals are even suggestive of positive welfare effects. Given this positive assessment, an open question is how investment tax incentives compare with other fiscal policies. I find that investment tax incentives change the level of production without improving productivity. While such a policy intervention may be cost-effective for compensating negative output shocks in the short to medium run, it may become increasingly costly over time compared to policies focused on long-term economic growth.

Another open question concerns the policy choice between investment tax credits and other tax incentives. I provide evidence that the refundability of investment tax credits can be an important feature for firms without taxable income. Nevertheless, the elasticity of investment is smaller than those found in studies of accelerated depreciation allowances. Preferably, both types of policies should be directly compared in

⁴³Chodorow-Reich (2019) summarizes employment multipliers from government spending and finds an average annual cost per job of about \$30,000. The distinct settings and features of these policies make a direct comparison with my estimate challenging. Importantly, with investment tax incentives, governments cover only a fraction of a firm's investment cost.

the same setting. Given the difficulty to find such settings, learning about the firms' perceptions of different incentives could help policymakers decide between different tax policies.

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